FitSense AI

Milestone: Implementation in MySQL

Group 3

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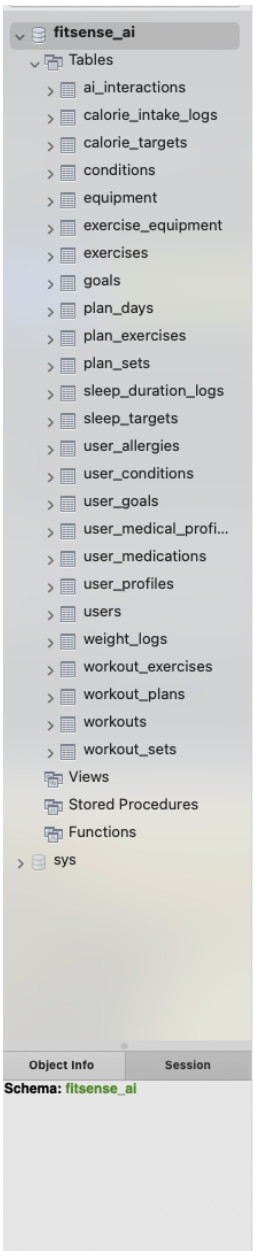
FitSense AI – Implementation in MySQL

Abhinav Kumar and Saisree Pothu

Attached Files:

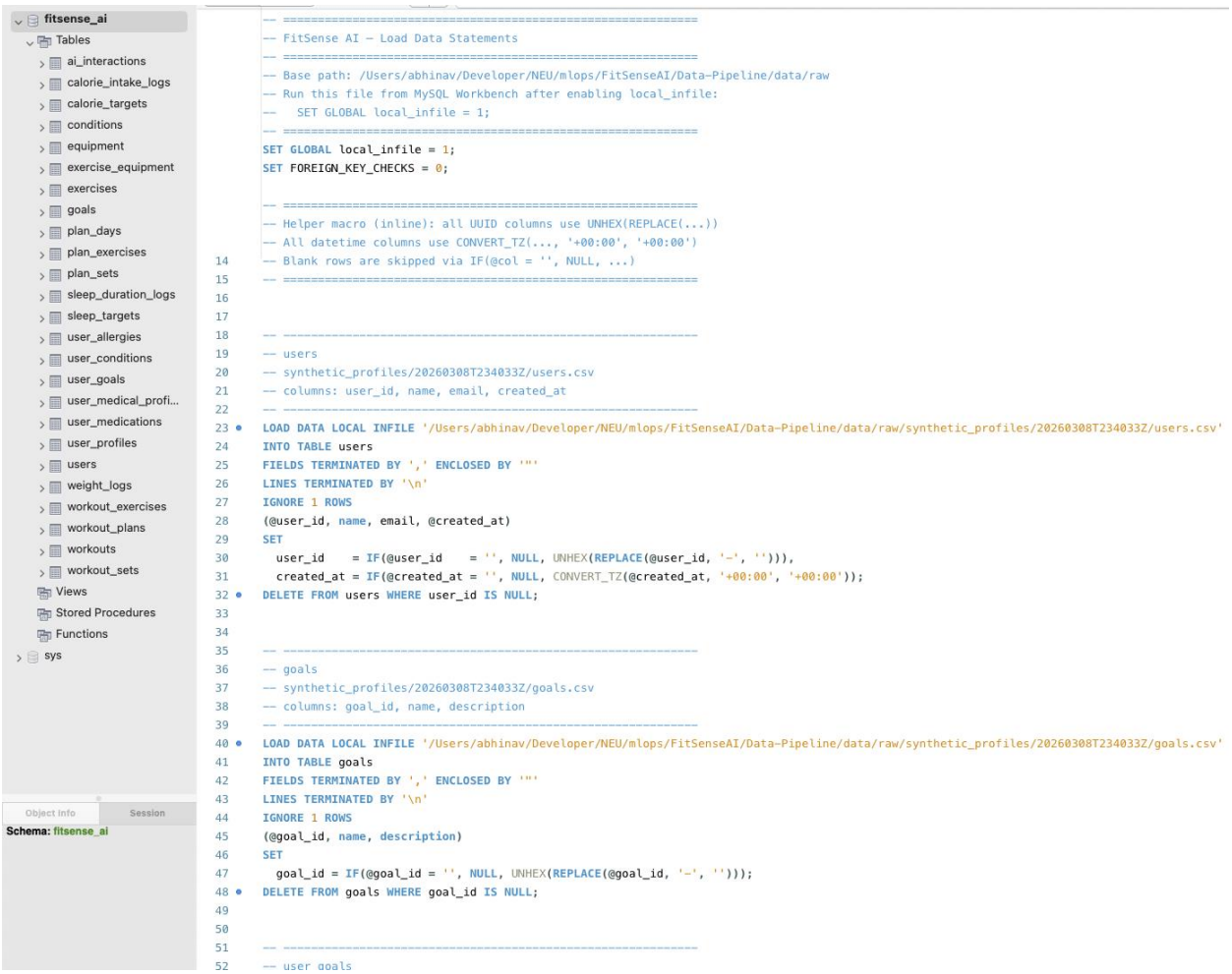
- **mysql.sql** - Contains all the sql statements to create database and all the tables
- **import_csv.sql** - The synthetic data was generated using python in csv format. The csv files were imported in the tables using these import statements
- **fitsense_ai_data_backup.sql** - Contains the data dump. It is a standalone single file that can be used to create the whole database with all the tables and all the data. It was prepared using mysqldump command
- **queries.sql** - Contains the sql statements for all the queries.

1. Database and table creation:




```
-- ============================================================
-- FitSense AI — MySQL Schema
-- ============================================================

CREATE DATABASE IF NOT EXISTS fitsense_ai;
USE fitsense_ai;

-- ============================================================
-- Users & Auth
-- ============================================================

SET FOREIGN_KEY_CHECKS = 0;

CREATE TABLE users (
  user_id    BINARY(16)   NOT NULL,
  name       VARCHAR(120) NOT NULL,
  email      VARCHAR(254) NOT NULL,
  created_at DATETIME(0)  NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (user_id),
  UNIQUE KEY uq_users_email (email)
);

-- ============================================================
-- Goals
-- ============================================================

CREATE TABLE goals (
  goal_id     BINARY(16)   NOT NULL,
  name        VARCHAR(120) NOT NULL,
  description TEXT,
  PRIMARY KEY (goal_id),
  UNIQUE KEY uq_goals_name (name)
);

CREATE TABLE user_goals (
  user_id  BINARY(16)  NOT NULL,
  goal_id  BINARY(16)  NOT NULL,
  priority TINYINT UNSIGNED,
  PRIMARY KEY (user_id, goal_id),
  KEY idx_user_goals_goal_id (goal_id),
  CONSTRAINT fk_user_goals_user FOREIGN KEY (user_id) REFERENCES users (user_id) ON DELETE CASCADE,
  CONSTRAINT fk_user_goals_goal FOREIGN KEY (goal_id) REFERENCES goals (goal_id) ON DELETE CASCADE
);

-- ============================================================
-- Conditions
-- ============================================================

CREATE TABLE conditions (
  condition_id BINARY(16)   NOT NULL,
  name         VARCHAR(120) NOT NULL,
  description  TEXT,
```

2. Importing synthetic data from csv



3. Queries

- **Basic Counts**

```sql
-- ============================================================
-- 1. BASIC COUNTS
-- ============================================================

-- Row counts for every table
SELECT 'users'                    AS tbl, COUNT(*) AS cnt FROM users
UNION ALL SELECT 'user_profiles',           COUNT(*) FROM user_profiles
UNION ALL SELECT 'user_goals',              COUNT(*) FROM user_goals
UNION ALL SELECT 'user_conditions',         COUNT(*) FROM user_conditions
UNION ALL SELECT 'user_allergies',          COUNT(*) FROM user_allergies
UNION ALL SELECT 'user_medications',        COUNT(*) FROM user_medications
UNION ALL SELECT 'user_medical_profiles',   COUNT(*) FROM user_medical_profiles
UNION ALL SELECT 'goals',                   COUNT(*) FROM goals
UNION ALL SELECT 'conditions',              COUNT(*) FROM conditions
UNION ALL SELECT 'equipment',               COUNT(*) FROM equipment
UNION ALL SELECT 'exercises',               COUNT(*) FROM exercises
UNION ALL SELECT 'workout_plans',           COUNT(*) FROM workout_plans
UNION ALL SELECT 'plan_days',               COUNT(*) FROM plan_days
UNION ALL SELECT 'plan_exercises',          COUNT(*) FROM plan_exercises
UNION ALL SELECT 'plan_sets',               COUNT(*) FROM plan_sets
UNION ALL SELECT 'workouts',                COUNT(*) FROM workouts
UNION ALL SELECT 'workout_exercises',       COUNT(*) FROM workout_exercises
UNION ALL SELECT 'workout_sets',            COUNT(*) FROM workout_sets
UNION ALL SELECT 'weight_logs',             COUNT(*) FROM weight_logs
UNION ALL SELECT 'sleep_duration_logs',     COUNT(*) FROM sleep_duration_logs
UNION ALL SELECT 'sleep_targets',           COUNT(*) FROM sleep_targets
UNION ALL SELECT 'calorie_intake_logs',     COUNT(*) FROM calorie_intake_logs
UNION ALL SELECT 'calorie_targets',         COUNT(*) FROM calorie_targets;
```

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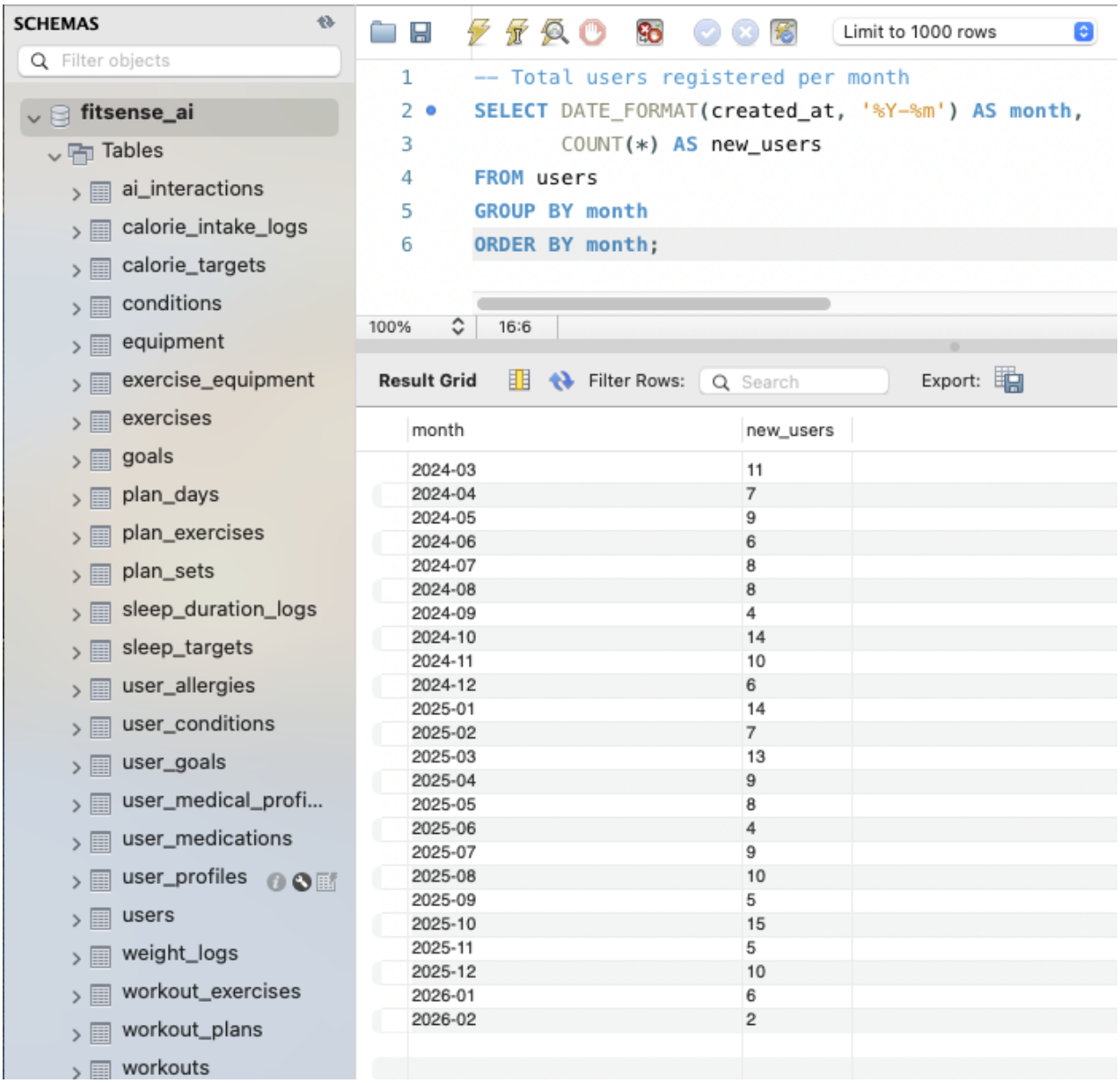
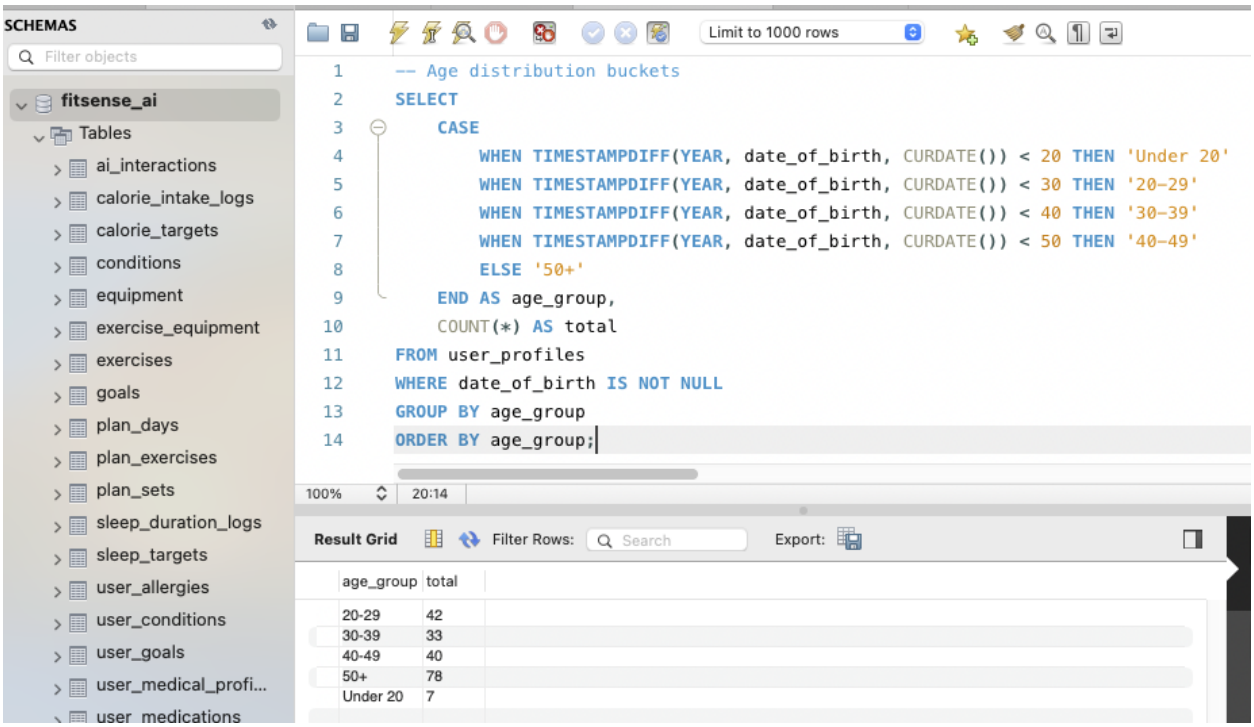
Result Grid Filter Rows: Search Export:

tbl	cnt
users	200
user_profiles	200
user_goals	297
user_conditions	198
user_allergies	187
user_medications	207
user_medical_profiles	200
goals	98
conditions	100
equipment	99
exercises	150
workout_plans	200
plan_days	811
plan_exercises	4044
plan_sets	12132
workouts	2400
workout_exercises	11963
workout_sets	35889
weight_logs	2600
sleep_duration_logs	12400
sleep_targets	200
calorie_intake_logs	12400
calorie_targets	200

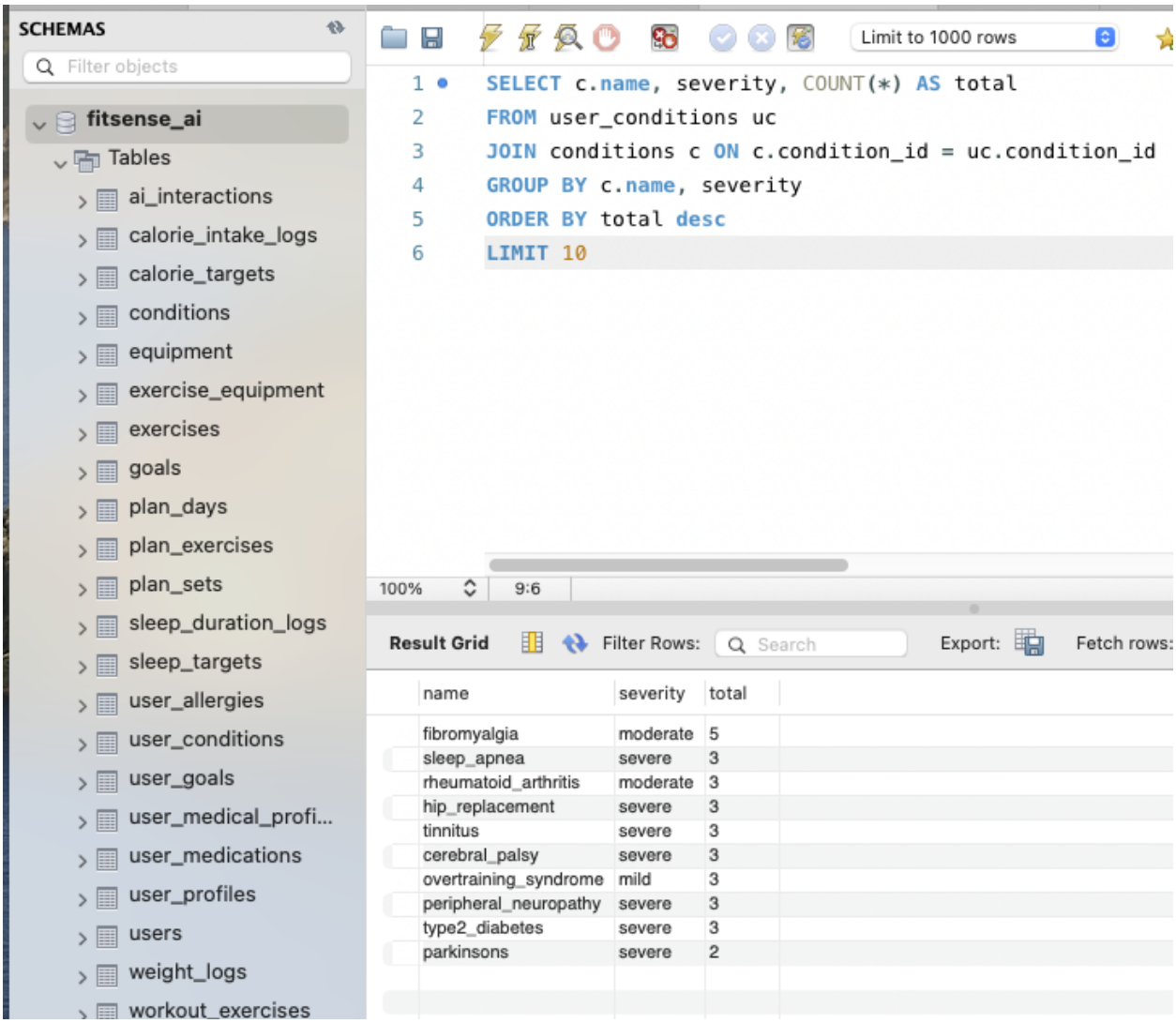
- **Total users registered per month**



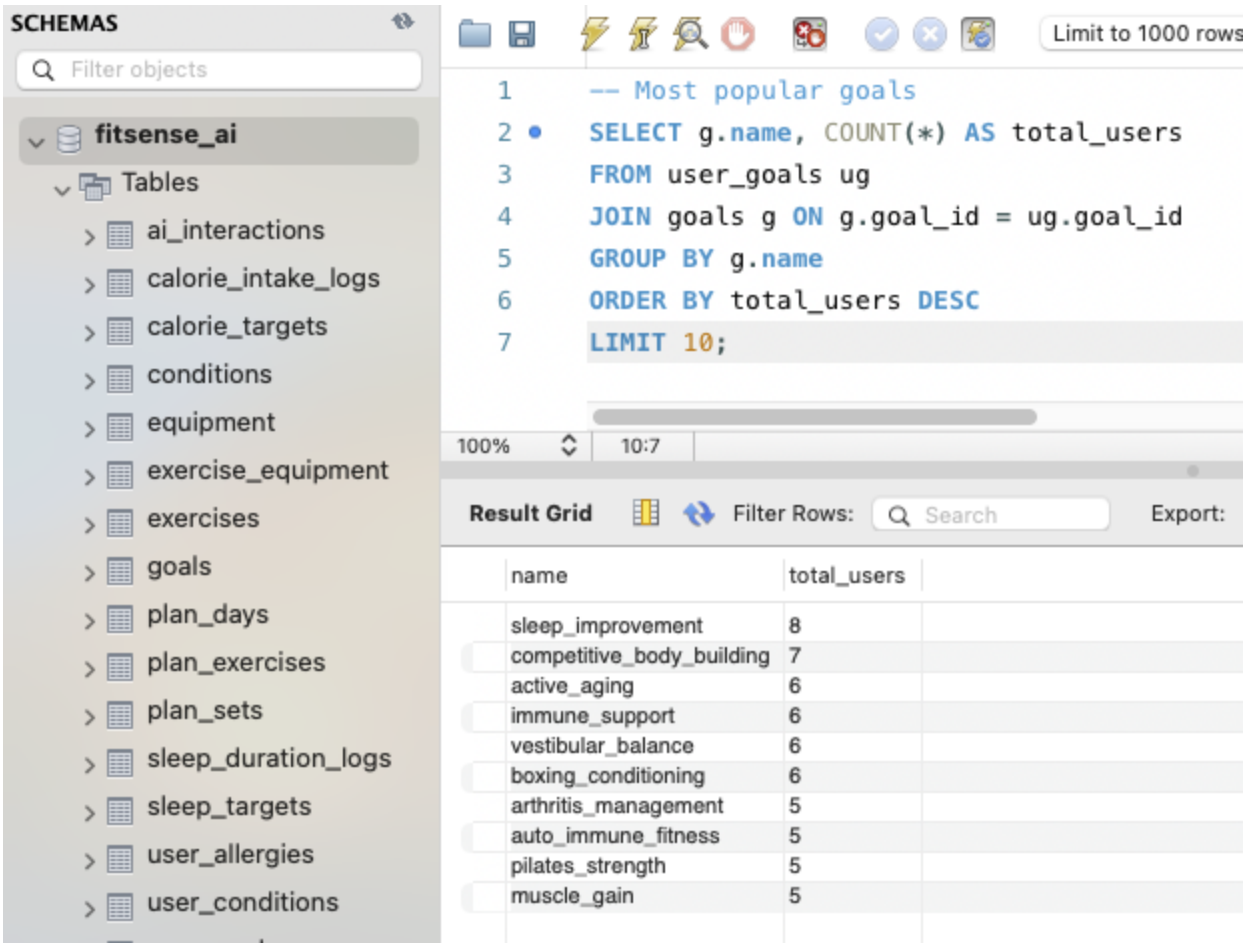
- **Age distribution buckets**



- **Most common conditions**



- **Most common goals**



- **Most common allergies**

SCHEMAS

Filter objects

- fitsense_ai
  - Tables
    - ai_interactions
    - calorie_intake_logs
    - calorie_targets
    - conditions
    - equipment
    - exercise_equipment
    - exercises
    - goals
    - plan_days
    - plan_exercises
    - plan_sets
    - sleep_duration_logs
    - sleep_targets
    - user_allergies
    - user_conditions

Limit to 1000 rows

```sql
-- Most common allergies
SELECT allergen, COUNT(*) AS total
FROM user_allergies
GROUP BY allergen
ORDER BY total DESC
LIMIT 10;
```

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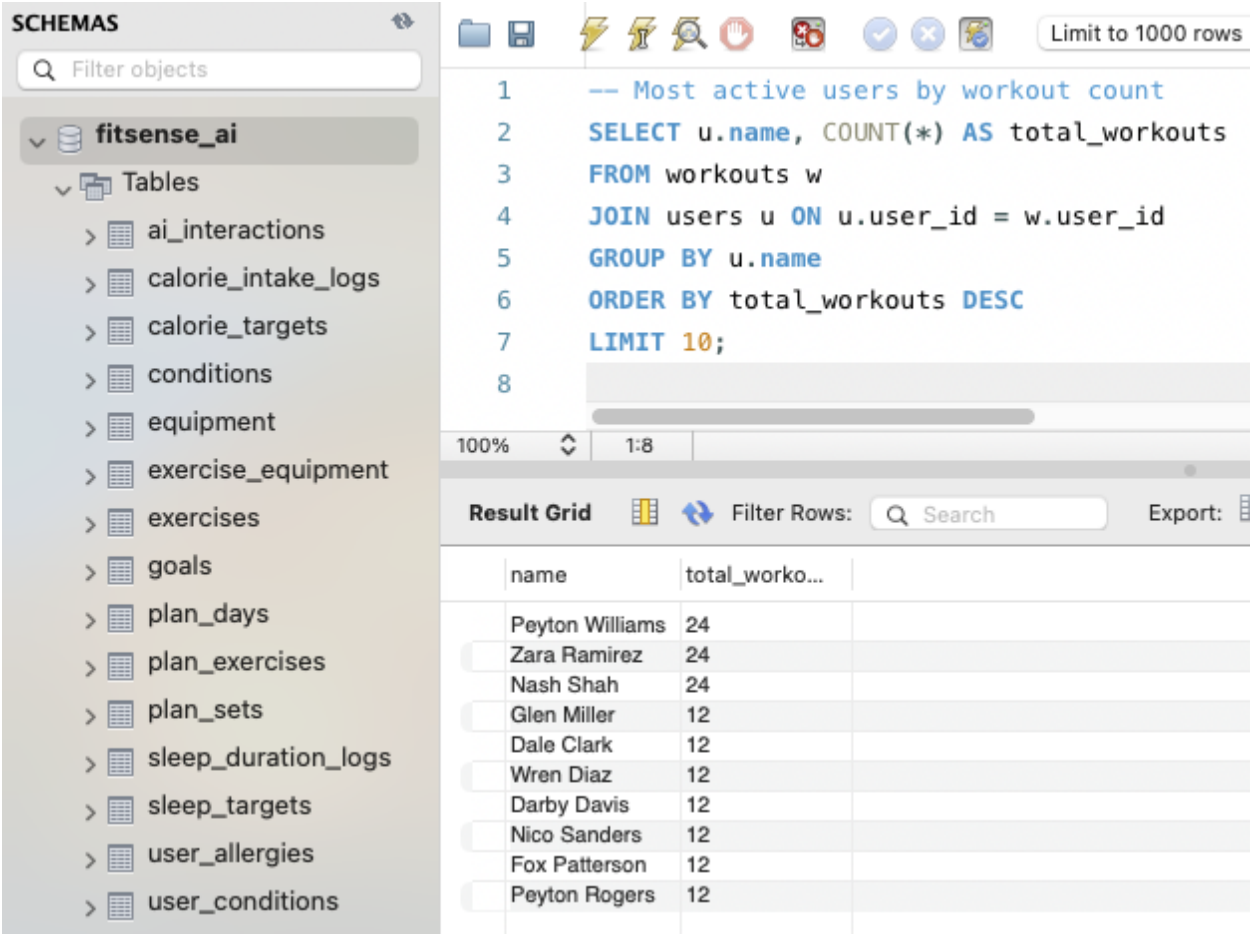
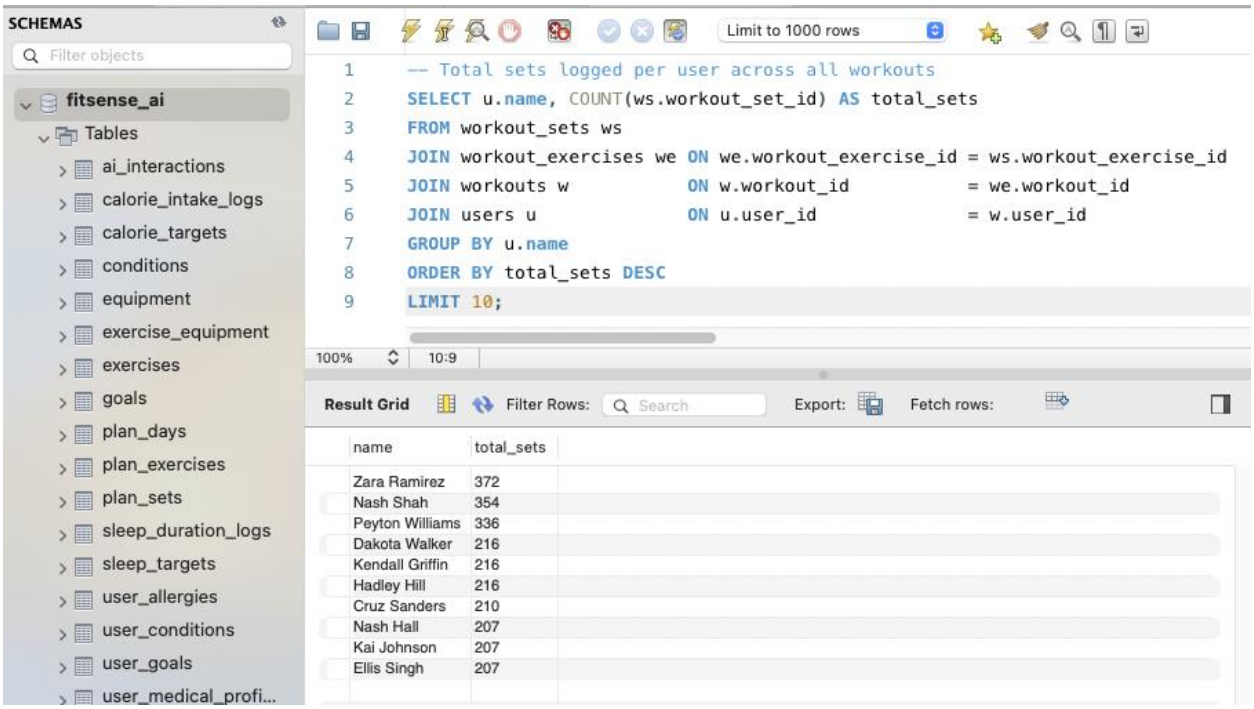
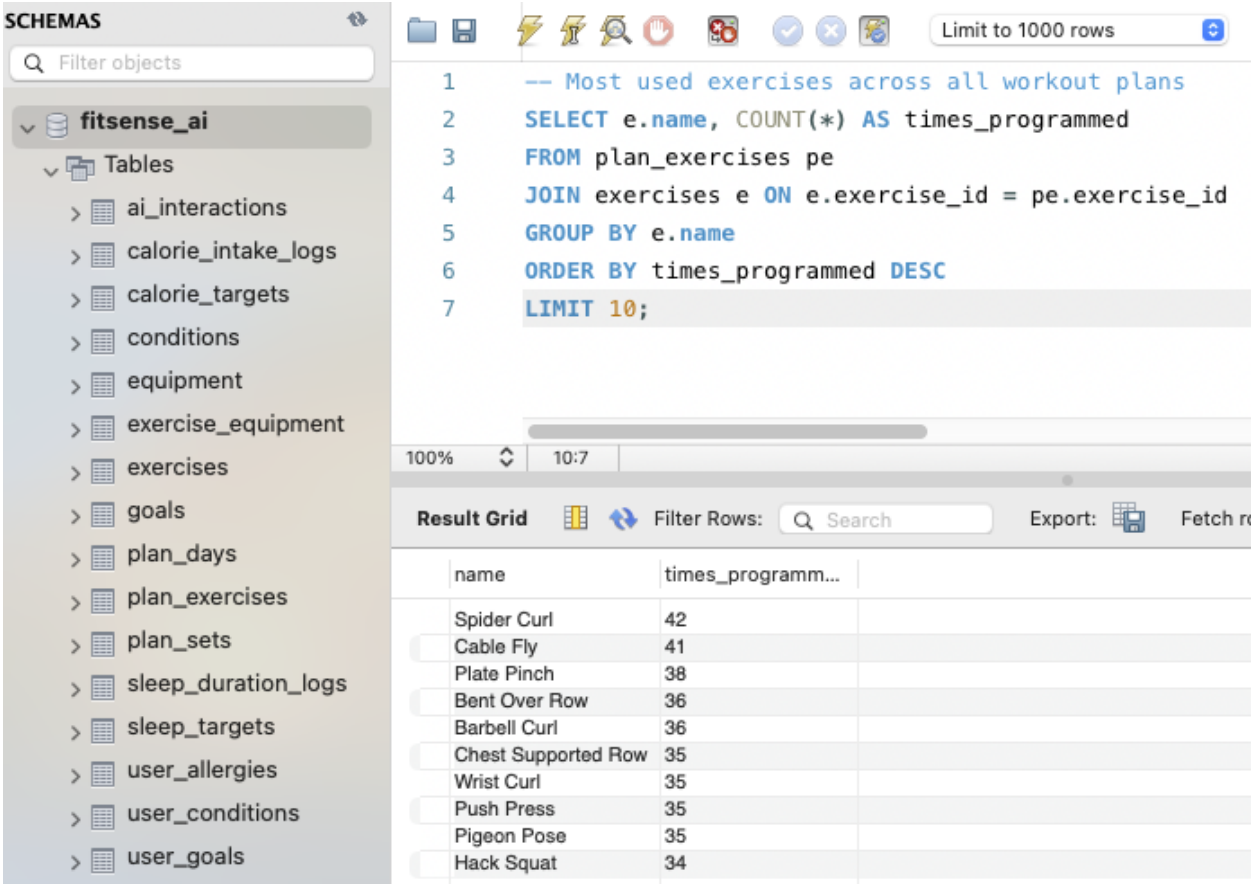
Result Grid Filter Rows: Search Export:

allergen	total
pine_nuts	7
dog_dander	5
avocado	5
dust_mites	4
dust	4
corn	4
hazelnuts	4
brazil_nuts	4
celery	4
cod	3

- **Most active users by workout count**



- **Total sets logged per user across all workouts.**



- **Most used exercises across all workout plans**